Statistical Thinking

Back to Basics

Learning Goals

- Define basic elements of a statistical investigation.
- Describe the role of p-values and confidence intervals in statistical inference.
- Describe the role of random sampling in generalizing conclusions from a sample to a population.
- Describe the role of random assignment in drawing cause-and-effect conclusions.
- Critique statistical studies.

Roadmap

Introduction

- Key Components to a Statistical Investigation

Distributional Thinking

Statistical Significance

- Control, Probability, Level of Significance

Generalizability

- Samples and Populations, Random Sample, Margin of Error

Cause and Effect

- Statistical Tendency, Random Assignment

Introduction

- 6 cups of coffee:
  - Men → 10% lower chance of dying
  - Women → 15% lower chance of dying
- Does this mean you should start drinking coffee or increase your own coffee habit?

Key Components to a Statistical Investigation

1. Planning the study
2. Examining the data
3. Inferring from the data
4. Drawing Conclusions

Roadmap

Introduction

- Key Components to a Statistical Investigation

Distributional Thinking

Statistical Significance

- Control, Probability, Level of Significance

Generalizability

- Samples and Populations, Random Sample, Margin of Error

Cause and Effect

- Statistical Tendency, Random Assignment

Distributional Thinking

- Analyzing the pattern of data variation, called the *distribution of the variable*, often reveals insights.
- Develop an example of a distribution, of one variable, that you often encounter in your life.
- What is the variable and how does it vary?

Table 1. Frequency tables of patient reading levels and pamphlet readability levels.

Patients' reading levels	< 3	3	4	5	6	7	8	9	10	11	12	> 12	Total
Count (number of patients)	6	4	4	3	3	2	6	5	4	7	2	17	63

Pamphlet's readability levels	6	7	8	9	10	11	12	13	14	15	16	Total
Count (number of pamphlets)	3	3	8	4	1	1	4	2	1	2	1	30

Distributional Thinking

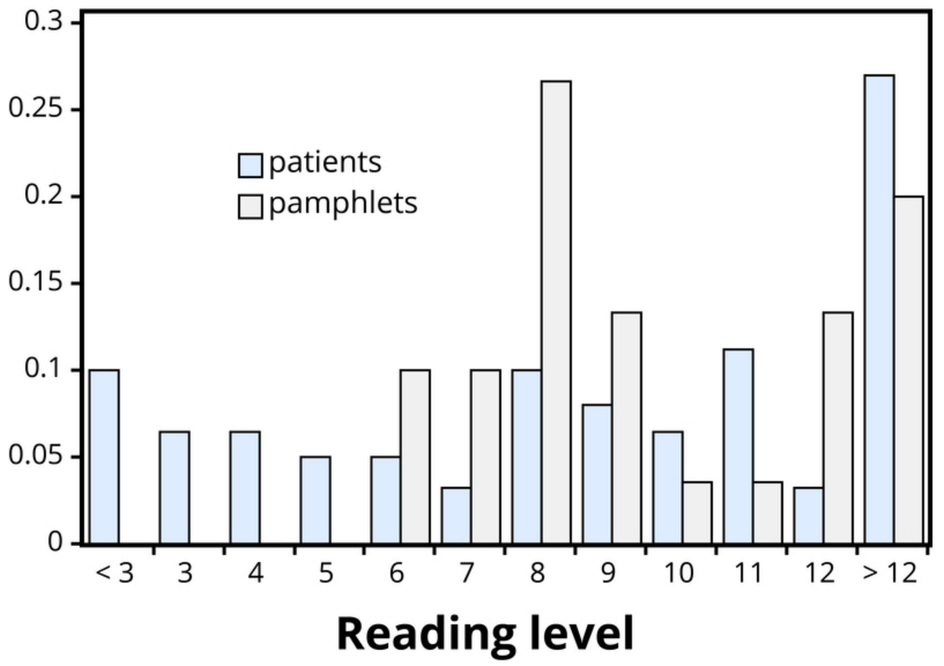


Figure 1. Comparison of patient reading levels and pamphlet readability levels.

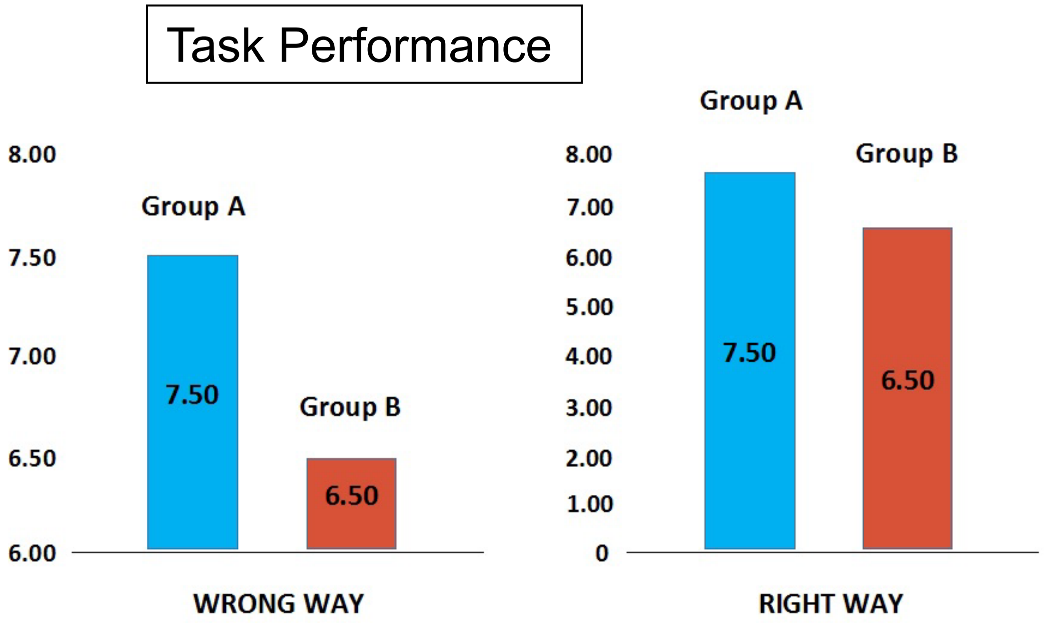
Collect and Display Data

- **How many concerts did you go to this past year?**
  - Write your answer down and turn it in

- **What is the best way to visually represent the information?**
- **What single score best represents the data?**
- **How much variability is in the data?**

Distributional Thinking

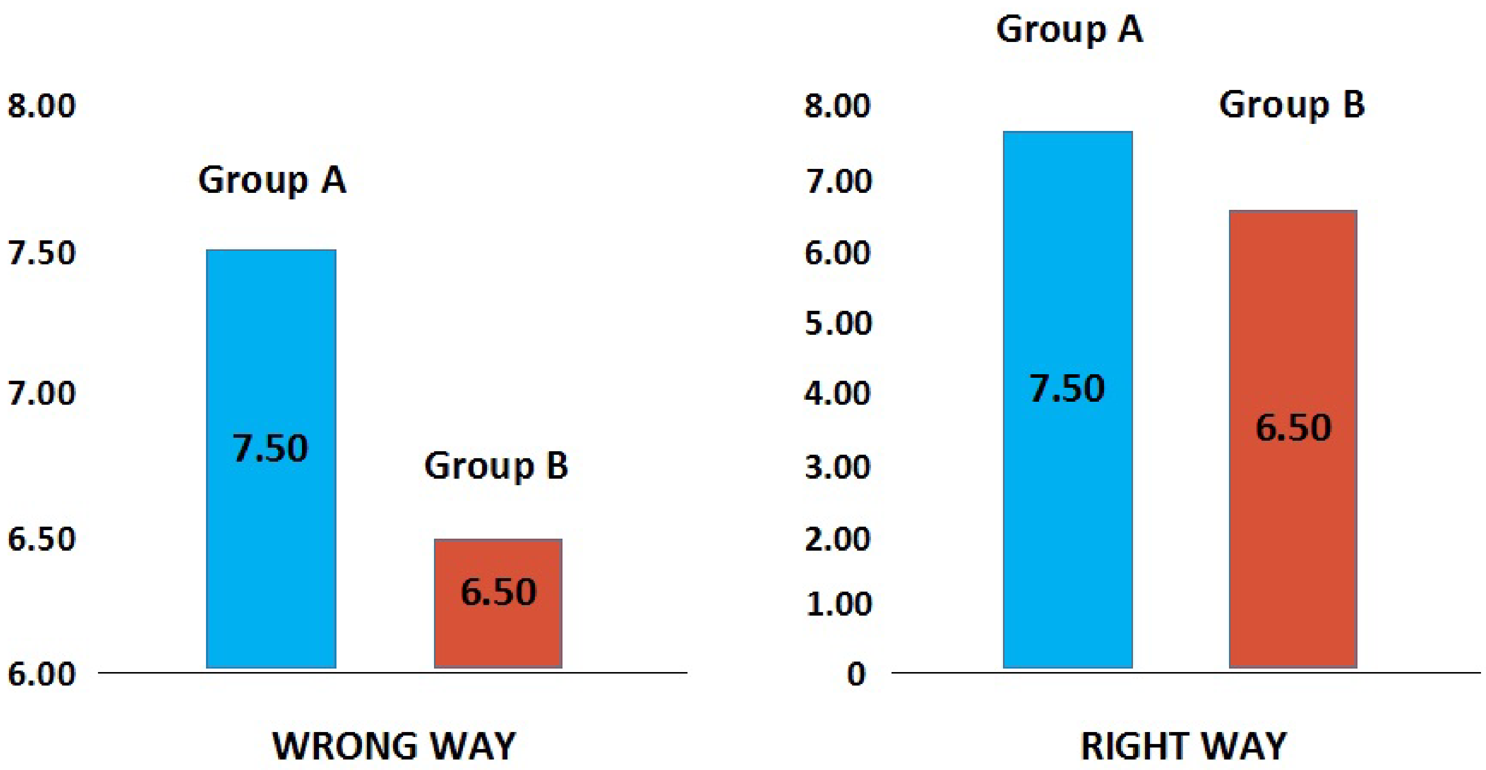
Tables and graphs are often presented in mass media; sometimes accurately, sometimes not.



1. What should you look for when examining this data to interpret it correctly?
2. How are people able to manipulate visual depictions of statistics to skew conclusions?

Distributional Thinking

- Tables and graphs are often presented in mass media; sometimes accurately, sometimes not.
    - What should you look for when examining this data to interpret it correctly?
    - How are people able to manipulate visual depictions of statistics to skew conclusions?



Roadmap

Introduction

- Key Components to a Statistical Investigation

Distributional Thinking

Statistical Significance

- Control, Probability, Level of Significance

Generalizability

- Samples and Populations, Random Sample, Margin of Error

Cause and Effect

- Statistical Tendency, Random Assignment

Statistical Significance

- **Control**
- Are there other variables in the infant study that the researchers missed?
- **Probability/ p-value**
  - [*P-Value extravaganza*]
- **Level of Significance**
  - $p < .05$

Roadmap

Introduction

- Key Components to a Statistical Investigation

Distributional Thinking

Statistical Significance

- Control, Probability, Level of Significance

Generalizability

- Samples and Populations, Random Sample, Margin of Error

Cause and Effect

- Statistical Tendency, Random Assignment

Generalizability: Samples and Populations

Generalizability

- How similar does a sample need to be to the population?
- Can you generalize from one class to the whole grade?
- Random Sample
- Margin of Error

Roadmap

Introduction

- Key Components to a Statistical Investigation

Distributional Thinking

Statistical Significance

- Control, Probability, Level of Significance

Generalizability

- Samples and Populations, Random Sample, Margin of Error

Cause and Effect

- Statistical Tendency, Random Assignment

Cause and Effect

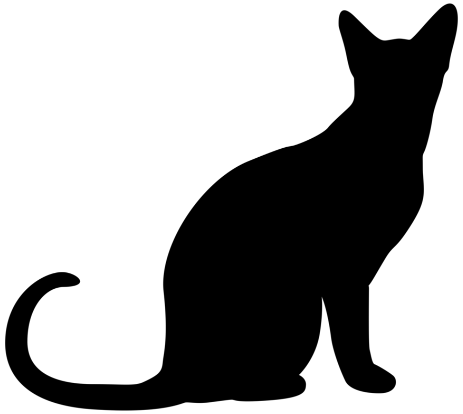
Statistical Tendency

Random Assignment

If the p-value is small (under .05), the observed results were (probably) not coincidental

CAT: The Muddiest Point

- What was the muddiest point about today's class?
- Write down what concept you are still struggling to understand.



Wrapping Up…